DSA REFERENCE

Greedy Failure Modes

Common wrong turns, debugging cues, and complexity traps.

Make the locally best irreversible choice only when an exchange or dominance argument proves an optimum still exists.

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DSA / GREEDY

DSA

Greedy Failure Modes

CORE IDEA

Make the locally best irreversible choice only when an exchange or dominance argument proves an optimum still exists.

Recognition cues

+ interval scheduling
+ minimum resources with ordering
+ locally dominant choice
+ matroid-like selection

Implementation sequence

+ identify candidate ordering
+ state local choice
+ give exchange proof
+ search for counterexamples
+ distinguish feasibility from optimality

Invariant

After each choice, there remains an optimal solution consistent with every choice already made.

Worked reasoning

Selecting the interval with earliest finishing time leaves the most room for all remaining intervals.

Complexity

Often O(n log n) for sorting plus O(n) scanning, with O(1) or O(n) auxiliary state.

Failure modes

- [ ] calling intuition a proof
- [ ] choosing earliest start instead of finish
- [ ] ignoring negative or zero cases
- [ ] using greedy where choices interact later

Practice ladder

Greedy Failure Modes: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Maximum Subarray**	State the cue, invariant, edge case, and proof before implementation.
2	**Jump Game**	State the cue, invariant, edge case, and proof before implementation.
3	**Gas Station**	State the cue, invariant, edge case, and proof before implementation.
4	**Partition Labels**	State the cue, invariant, edge case, and proof before implementation.
5	**Non-overlapping Intervals**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

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